THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 11 & 12: **Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1 C/s	$I = Q/t$
2	**(c)** in series	Same current
3	**(d)** 6 Ω	$150/25$
4	**(a)** Correct	Single path
5	**(b)** 2 kWh	0.5×4
6	**(c)** the north pole of a compass points	Tangent to field line
7	**(b)** poles	Stronger field
8	**(d)** Oersted	1820
9	**(c)** Incorrect — they never intersect	Two directions impossible
10	**(b)** inversely proportional to r	Weakens with distance

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Property that opposes the flow of current, $R = V/I$; ohm (Ω).	1+1
12	$W = QV = 30$ J	2
13	Concentric circles in a horizontal plane; anticlockwise when viewed from above.	2
14	The magnetic field of the conductor interacts with the external magnetic field.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $R = 9$ Ω, $I = 1$ A (ii) $R = 2$ Ω, $I = 4.5$ A	1½+1½
16	Sprinkle iron filings around a solenoid on cardboard; pattern resembles bar magnet; one end acts as N, other as S (compass test)	3
17	Square of current, resistance, time: $H = I^2Rt$	2+1
	OR: $P = 1100$ W; $E = 2.2$ kWh $= 7.92 \times 10^6$ J	1+2
18	Current (directly), number of turns (directly), radius (inversely)	3
	OR: Safety device of a low-melting wire in series with the live wire; excessive current heats and melts it, breaking the circuit	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $V \propto I$ at constant temperature; not valid for non-ohmic conductors/changing temperature (ii) straight line through origin; $R \approx 16.6$ Ω (slope)	2+3
	OR: (i) 2000 + 300 + 100 = 2400 W (ii) $I = 2400/220 = 10.9$ A (iii) 15 A (iv) $2.4 \times 3 = 7.2$ kWh (v) each gets 220 V; independent working	1+1+1+1+1
20	(i) imaginary lines showing direction and strength of field (ii) place compass near N pole, mark its ends, move it repeatedly, join points (iii) emerge N, enter S; closed curves; never intersect	1+2+2
	OR: (i) towards the west (ii) towards the east (iii) Oersted (iv) current produces a magnetic field (v) by increasing current	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 11 & 12: **Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Student A is correct	Must melt
2	**(a)** nature of material and temperature	Not dimensions
3	**(d)** double	Ohm's law
4	**(c)** four times	I^2
5	**(b)** rheostat	Variable resistor
6	**(b)** Student A is correct	With live wire
7	**(b)** black	Live red
8	**(a)** heating effect of current	Melts
9	**(b)** the same at all points	Uniform
10	**(a)** reverses	Rule

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Low resistivity; good conductor; ductile.	2
12	False — resistance is directly proportional to length.	2
13	Increase the current; increase the number of turns (use soft iron core).	1+1
14	False — it carries current only when there is leakage.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) 3 Ω and 6 Ω in parallel (2 Ω), in series with 2 Ω (ii) all three in parallel	1½+1½
16	It experiences a force perpendicular to both field and velocity, so its path bends (circular); Fleming's left-hand rule with current opposite to electron motion	3
17	If one appliance fails, circuit breaks; all get the same current regardless of need; voltage divides, so devices don't get full voltage	3
	OR: $R = 20\ \Omega$; $I = 25/20 = 1.25$ A	1½+1½
18	Metal body connected to earth via green wire; leakage current flows to earth, keeping the body at zero potential, preventing shock	3
	OR: Each segment produces circular field lines; near the centre arcs become straight and add up in one direction	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $R \propto l$, $R \propto 1/A \Rightarrow R = \rho l/A$ (ii) $l = RA/\rho = 10 \times 2 \times 10^{-7}/10^{-6} = 2$ m	2+3
	OR: (i) $P = VI$, kWh $= 3.6 \times 10^6$ J (ii) daily $1.2 + 0.6 = 1.8$ kWh; monthly 54 units; ₹270	2+3
20	(i) live (220 V), neutral (0 V) and earth; parallel connections; separate circuits of 15 A and 5 A (ii) fuse, MCB, earthing, proper insulation	3+2
	OR: (i) like bar magnet, uniform inside (ii) soft iron core inside a current-carrying solenoid (iii) clockwise current seen from one end → S pole; anticlockwise → N pole	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 11 & 12: **Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** watt	J/s
2	**(a)** Correct	Alloy
3	**(b)** 12 Ω	$R \propto l$
4	**(c)** 1.6×10^{-19} C	Negative
5	**(a)** Student A is correct	Add
6	**(a)** compass	Needle
7	**(a)** Correct	Angle 0°
8	**(d)** in parallel	Same voltage
9	**(b)** 10 A	$I \approx 9$ A
10	**(a)** Student A is correct	Smaller radius → stronger

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$I = P/V = 2000/220 = 9.09$ A	2
12	It melts and breaks the circuit when the current exceeds a safe value.	2
13	It switches off the circuit automatically when the current exceeds a safe limit.	2
14	A stronger magnetic field in that region.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$R \propto 1/A \Rightarrow$ 10 Ω	3
16	Red/brown, black/blue, green; so that the appliance is disconnected from the high potential when switched off	2+1
17	(i) $I = 2$ A, $P = I^2R = 8$ W (ii) $V = 4$ V across 2 Ω, $P = V^2/R = 8$ W; equal	1½+1½
	OR: Electric iron, toaster, heater, electric bulb, fuse	3
18	(i) increases (ii) reverses (iii) becomes zero	1+1+1
	OR: Short circuit: live and neutral touch, resistance falls, huge current; overloading: too many appliances drawing more current than rated	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) B (ii) $R \propto l/A$: $(2/0.5)/(1/1) = 4$ (iii) resistance of a unit cube, ρ (iv) same material and temperature (v) silver/copper	1+1+1+1+1
	OR: (i) $R = 5/2.5 = 2$ Ω; $10/n = 2 \Rightarrow n = 5$ (ii) two in series (8 Ω) in parallel with two in series (8 Ω) gives 4 Ω	3+2
20	(i) force on a current-carrying conductor (ii) towards south (iii) direction remains the same (iv) Fleming's left-hand rule (v) light and good conductor, non-magnetic	1+1+1+1+1
	OR: (i) thumb along current, fingers give field (ii) out of the plane towards the viewer (iii) straight and perpendicular to the plane	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 11 & 12: **Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 40 W	I^2R
2	**(c)** opposite to the flow of electrons	Positive to negative
3	**(b)** $R/4$	Each $R/2$, in parallel
4	**(c)** volt	J/C
5	**(c)** 3 units	1.5×2
6	**(b)** temporary magnet	Electromagnet
7	**(a)** long current-carrying solenoid	Inside
8	**(a)** the direction of current is reversed	Also if field reversed
9	**(c)** a sudden large current	Very low resistance
10	**(c)** leakage of current (faulty earthing)	Earth fault

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3/3 = 1\ \Omega$	2
12	Resistance of a conductor through which 1 A flows when a potential difference of 1 V is applied.	2
13	Copper has low resistance and is non-magnetic.	2
14	220 V; neutral is at (nearly) zero potential.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$5 \times 40 \times 6 \times 30 = 36\,000$ Wh $= 36$ kWh; cost = ₹144	2+1
16	Current produces a magnetic field which acts on the needle; when switched off, needle returns to N–S	3
17	Ammeter must carry the full current (low resistance); voltmeter measures potential difference between two points (high resistance), drawing negligible current	3
	OR: $\frac{1}{R} = \frac{1}{100} + \frac{1}{50} + \frac{1}{500} = \frac{16}{500} \Rightarrow R = 31.25\ \Omega$; current = 7.04 A	2+1
18	Electric bell, cranes for lifting iron, MRI/relays, loudspeakers	3
	OR: Heaters draw larger currents; separate circuits with suitable fuse ratings prevent overloading	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) high resistance tungsten heats to high temperature and emits light (ii) to prolong filament life (iii) $H = Pt = 60 \times 300 = 18\,000$ J	2+1+2
	OR: (i) work done per unit charge between two points (ii) 6 J (iii) $1/(1.6 \times 10^{-19}) = 6.25 \times 10^{18}$	1+2+2
20	(i) as defined (ii) melts and breaks the circuit when current exceeds the rating (iii) $I = 4.5$ A, so 5 A fuse	2+1+2
	OR: (i) current produces a magnetic field around it (ii) concentric circles near wire, straight at centre (iii) electromagnets, electric bell, loudspeaker	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.